CFEV Assignment 2: Which project to Invest in ?

Q1. Can you rank these projects simply by inspecting the cash flows?

No.
Projects cannot be reliably ranked by simple inspection of cash flows because:

- Cash flows occur at different points in time, the cash in year 3 is worth less than same cash amount in year 4
- Some projects have late lump-sum inflows (e.g., Project 3), you invest in year 1 but get return at year 10
- Some have negative terminal cash flows (e.g., Project 4), at the end of project you have to give out more

Simple inspection may give misleading rankings, especially when timing differs significantly.

Q2. What criteria will you use to rank the projects? Which quantitative ranking methods are better? Why?

Primary criterion: Net Present Value (NPV)

Why NPV is the best method:

- Time value of money is considered
- Considers all cash flows
- Measures with discount rate always

Other methods (secondary):

- IRR – can be misleading with non-conventional cash flows
- Payback period – ignores time value and later cash flows

Q3. What ranking of the projects did you find using the quantitative method? Does it differ from simple inspection?

Ranking based on NPV (10% discount rate):

1. Project 3
2. Project 4
3. Project 8
4. Project 7
5. Project 5
6. Project 1
7. Project 6
8. Project 2

It differs from simple inspection as :

- Simple inspection would favour Project 3 only due to high total cash
- It would undervalue Project 4 because of its terminal loss
- It would overlook early-cash projects like Project 7

Q4. What kind of real investment projects have cash flows similar to those in Exhibit 1?

Projects with similar cash-flow patterns include:

1. Infrastructure projects, Roads, bridges, power plants (steady inflows + decommissioning)
2. Manufacturing plants (Equipment replacement or dismantling costs at end of life)
3. Technology or R&D projects (Long gestation periods followed by payoff)

These projects often involve:

- Early operating cash flows
- Large terminal costs or benefits
- Non-conventional cash-flow patterns